Presenter Guide for the Slides on Neyman Orthogonality

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1 Symbols and how to read them

1.1 Conventions

- A subscript 0 marks the true value: θ_0 is read "theta zero" or "theta naught", g_0 "g zero".
- A hat marks an estimate: $\hat{\theta}$ is read "theta hat", $\hat{\eta}_k$ "eta hat k", the nuisance estimated without fold k.
- A subscript ξ marks a quantity of a simulated data-generating process in pretraining: g_ξ is "g of xi", the outcome regression of the simulated process ξ.
- A subscript Π marks a posterior mean under the prior Π: $\hat{g}_\Pi$ is "g hat pi", the posterior mean of the outcome regression.
- Letters in calligraphic font are sets or classes: $\mathcal{T}_N$ "script T N", $\mathcal{H}$ "script H", $\mathcal{D}_n$ "script D n" for the data set.

1.2 Greek letters

Symbol	Name	Meaning on the slides
θ, θ_0, $\hat{\theta}$	theta	target parameter, its true value, its estimate; $\theta(g)$ is the target computed at a regression g
η, η_0	eta	nuisance (all unknown functions besides the target), and its true value
α, α_0	alpha	Riesz representer; for the ATE $\alpha_0 = D/m_0 - (1-D)/(1-m_0)$, the inverse propensity weights
ν	nu	the map that defines the target, $\theta(g) = \mathbb{E}[\nu(W; g)]$; for the ATE $\nu(W; g) = g(1, X) - g(0, X)$
ψ	psi	score, the function inside the estimating equation $\mathbb{E}[\psi] = 0$; ψ_{DR} is the AIPW score, ψ_{IV} and ψ_{PO} the scores of the partially linear model
φ, φ_ξ	phi (curly)	efficient influence function (EIF), or a gradient
ε	epsilon	outcome noise in the partially linear model

Symbol	Name	Meaning on the slides
ρ	rho	residual in a conditional moment restriction $\mathbb{E}[\rho \mid Z] = 0$
ω, ϱ	omega, rho (curly)	weight density of the average derivative, and conditional density of D given X
δ	delta	size of a shift of the outcome regression, $Y \mapsto Y + \delta h(Z)$
δ_N, Δ_N, λ_N	delta, Delta, lambda	sequences tending to zero in the conditions of the DML theorem; λ_N bounds the derivative under near-orthogonality
λ	lambda	ridge penalty in the Gaussian simulation (appendix)
ξ	xi	a data-generating process drawn from the pretraining prior
Π	capital pi	prior over data-generating processes
κ	kappa	coefficient of the shift along the Riesz representer in the prior correction, $g_\xi + \kappa\alpha_\xi$
σ_n^2, σ^2	sigma squared	variance of κ in the prior correction; asymptotic variance in the DML theorem
ϖ_i	pi (curly, "varpi")	Bayesian bootstrap weights in the label correction
χ	chi	parameters of a trained network F_χ
τ, τ_ξ	tau	conditional average treatment effect function $g(1,x) - g(0,x)$; in F_τ the mixing weight of a path
γ, β	gamma, beta	coefficients of the treatment and outcome equations in the Gaussian simulation

1.3 Other symbols

Symbol	Read as	Meaning
$W = (Y, Z)$, $Z = (D, X)$		observation: outcome Y, regressors Z, treatment D, covariates X
$g_0(z) = \mathbb{E}[Y \mid Z = z]$	"g zero of z"	outcome regression
m_0, ℓ_0, f_0	"m zero", "ell zero", "f zero"	propensity score or treatment regression $\mathbb{E}[D \mid X]$; $\mathbb{E}[Y \mid X]$; nonparametric part of the partially linear model
V		treatment residual $D - m_0(X)$
$\mathbb{E}$, $\mathbb{E}_P$	"expectation (under P)"	population mean
$\mathbb{E}[\cdot \mid Z]$	"given Z"	conditional expectation
$\mathbb{P}_n$, $\mathbb{E}_{n,k}$	"P n", "E n k"	average over the whole sample, average over fold I_k
$\langle f, g \rangle$	"inner product of f and g"	$\mathbb{E}[f(Z)g(Z)]$
$\|f\|$	"norm of f"	$\sqrt{\mathbb{E}[f(Z)^2]}$, the root mean squared size
$L^2(P_Z)$	"L two of P Z"	functions of Z with finite mean square
$M(\theta, \eta)$	"moment map"	$\mathbb{E}_P[\psi(W; \theta, \eta)]$, the expected score at a fixed nuisance
J_0	"J zero", the Jacobian	derivative of M in θ at the truth
∂_r, ∂_δ	"partial derivative in r"	derivative in a scalar parameter
$\mathrm{D}_r[\eta - \eta_0]$	"D r applied to eta minus eta zero"	Gateaux derivative of the moment map in the direction $\eta - \eta_0$; square brackets mark the direction
$\mathcal{T}_N$	"script T N"	nuisance realisation set, which contains the nuisance estimates with high probability
T_1, T_2, T_3	"T one" etc.	the three terms of the error decomposition
$O_P(a_n)$, $o_P(a_n)$	"big O P", "little o P"	bounded in probability by a multiple of a_n; smaller than a_n in probability

Symbol	Read as	Meaning
$\rightsquigarrow N(0, \sigma^2)$	"converges in distribution to a normal"	asymptotic normality
$\propto$	"proportional to"	equal up to a constant factor
$\otimes$	"product"	product of distributions, the mechanisms combined
$\perp\!\!\!\perp$	"independent of"	independence
$\sim$	"distributed as"	e.g. $\kappa \sim N(0, \sigma_n^2)$
$\mapsto$	"maps to"	e.g. $\delta \mapsto b_n(\delta; h)$, the bias as a function of δ
$:=$	"is defined as"	definition
$P_{\delta h}$	"P delta h"	law of the data after the outcomes are shifted by $\delta h(Z)$
$b_n(\delta; h)$	"b n of delta and h"	bias of the estimator at sample size n along this shift
$w_i = n\,\partial\hat{\theta}/\partial Y_i$	"implicit weight of observation i"	how much the estimate moves when outcome i moves, times n
$\mathcal{D}_n$	"script D n"	data set of n observations, the context of an in-context learner
F_χ, a_χ	"F chi", "a chi"	trained network, its amortisation error

1.4 How to read the main equations

1. $\theta(g) = \langle \alpha_0, g \rangle$ and $\theta(g_0 + h) - \theta_0 = \langle \alpha_0, h \rangle$.
   "The target at any regression g is the weighted average of g with weights α_0. If the regression is wrong by h, the target is wrong by the α_0-weighted average of h." This is why α_0 is the derivative of the target with respect to the regression.

2. $\psi_{\mathrm{DR}}(W; \theta, g, \alpha) = \nu(W; g) - \theta + \alpha(Z)\{Y - g(Z)\}$.
   "Plug-in estimate minus the target, plus the residual weighted by the Riesz representer." For the ATE: $g(1, X) - g(0, X) - \theta + \alpha(Z)\{Y - g(Z)\}$, the AIPW score.

3. $\frac{1}{n} \sum_k \sum_{i \in I_k} \psi(W_i; \theta, \hat{\eta}_k) = 0$.
   "Solve for θ the sample average of the score, where each observation uses nuisances fitted without its own fold." This is DML with cross-fitting.

4. $\mathbb{E}_{n,k}\psi(W; \theta_0, \hat{\eta}_k) = T_1 + T_2 + T_3$.
   "The score at the estimated nuisance splits into noise at the true nuisance (T_1), the fluctuation of the difference (T_2), and the bias from the nuisance error (T_3)." $T_3 = M(\theta_0, \hat{\eta}_k)$ is the expected score at the estimated nuisance.

5. $\mathrm{D}_r[\eta - \eta_0] = \partial_r \mathbb{E}_P[\psi(W; \theta_0, \eta_0 + r(\eta - \eta_0))]$ and $\mathrm{D}_0[\eta - \eta_0] = 0$.
   "Move the nuisance from the truth a fraction r of the way towards η and differentiate the expected score in r. Orthogonality says that this slope is zero at $r = 0$ for every admissible η." In the figure it is the slope of each curve at the origin.

6. $\mathrm{D}_0[h] = \langle \alpha_0 - A, h \rangle$.
   "With correction weights A, the slope in direction h is the gap between α_0 and A, averaged against h." It vanishes for all h only when $A = \alpha_0$.

7. $M(\theta_0, g, \alpha) = -\langle g - g_0, \alpha - \alpha_0 \rangle$.
   "The bias of the AIPW score is minus the product of the two nuisance errors." This is the mixed bias property; the bias is second order and vanishes if either error is zero.

8. $M_{\mathrm{PO}} = \langle \ell - \ell_0, m - m_0 \rangle - \theta_0 \|m - m_0\|^2$.
   "The partialling-out bias contains a square of the treatment error." The score is orthogonal, but

not doubly robust when $\theta_0 \neq 0$.

9. $\mathbb{E}[\alpha(Z)^2 - 2\nu(W;\alpha)] = \|\alpha - \alpha_0\|^2 - \|\alpha_0\|^2$.
"The Riesz loss equals the squared distance to the true representer, up to a constant." Minimising it learns α_0 without a formula for it.

10. $\sqrt{N}(\hat{\theta} - \theta_0) = -J_0^{-1} N^{-1/2} \sum_i \psi(W_i; \theta_0, \eta_0) + o_P(1) \rightsquigarrow N(0, \sigma^2)$ (appendix).
"The scaled error behaves like an average of the score at the true nuisance, so it is asymptotically normal."

11. $\theta_\xi + \sum_i \varpi_i \varphi_\xi(W_i)$.
"The training label of the label correction: the true target of the simulated process plus a randomly weighted average of its efficient influence function on the simulated data."

12. $g_\xi + \kappa \alpha_\xi$ with $\kappa \sim N(0, \sigma_n^2)$.
"The prior correction shifts each simulated outcome regression along its own Riesz representer by a random amount."

13. $P_{\delta h}$, $b_n(\delta; h) = \mathbb{E}_{\delta h}[\hat{\theta}] - \theta(P_{\delta h})$, $\partial_\delta b_n(0; h) = 0$.
"Shift all outcomes by $\delta h(Z)$; the bias of the estimator after the shift is b_n. First-order insensitivity says the bias does not change to first order in δ."

14. $\Pi(\mathrm{d}\xi_X, \mathrm{d}\xi_Y \mid \mathcal{D}_n) \propto \Pi_{X,Y}(\mathrm{d}\xi_X, \mathrm{d}\xi_Y) \prod_i p(X_i; \xi_X)\, p(Y_i \mid D_i, X_i; \xi_Y)$.
"Under independent mechanisms, the posterior of the covariate and outcome mechanisms is the prior times the covariate and outcome likelihood. The treatment likelihood does not appear."

15. $\partial_\delta b_n(0; h) = \mathbb{E}_P[n^{-1} \sum_i w_i h(Z_i)] - \langle \alpha_0, h \rangle$ (appendix).
"The sensitivity of an estimator equals the expected average of its implicit weights times h, minus the true change of the target." The estimator is insensitive when its implicit weights act like the Riesz representer.

2 Speaker notes

Title slide

- Three parts: what Neyman orthogonality is, how the literature generalises it, and a project on estimators without a score.
- Notation follows the draft of the planned paper, close to Chernozhukov et al. (2018) and Chernozhukov, Newey and Singh (2022).

Data, target and Riesz representer

- All targets are linear functionals of the outcome regression g_0; the ATE is the running example.
- The Riesz representer α_0 is the derivative of the target in the regression; for the ATE it is the inverse propensity weight.
- Errors in g move the target by $\langle \alpha_0, h \rangle$, the source of plug-in bias.

Scores, moment conditions and cross-fitting

- A score turns the target into an estimating equation $\mathbb{E}[\psi] = 0$; three examples, AIPW and the two partially linear scores.
- The choice of instrument in a conditional moment restriction decides sensitivity and efficiency.
- DML solves the cross-fitted sample equation, so each observation uses nuisances fitted without it.

Three-term decomposition

- T_1 is the oracle noise, T_2 the overfitting term, T_3 the nuisance bias.
- Each term has its own remedy: CLT, cross-fitting, orthogonality and rates.
- For the plug-in score $T_3 = \langle \alpha_0, \hat{g} - g_0 \rangle$ exactly, which is too large for machine learning rates.

Definition

- Orthogonality: the Gateaux derivative of the expected score in the nuisance vanishes at the truth.
- It is a joint property of score, law and the set $\mathcal{T}_N$.
- Near-orthogonality allows a derivative of order $o(N^{-1/2})$.

Moment maps along a nuisance direction

- Each curve is the expected score as the nuisance moves away from the truth.
- Orthogonal scores have zero slope at $r = 0$: AIPW in (a), both partially linear scores in (b).
- Plug-in and IPW in (a) and the instrument D in (b) have a nonzero slope.

Construction of orthogonal scores

- For linear functionals the only orthogonal correction is the Riesz representer times the residual.
- In the partially linear model any instrument with $\mathbb{E}[A \mid X] = 0$ works, so orthogonal scores are not unique.
- Historically, Neyman's residual score projects off the nuisance scores.

Double robustness and the mixed bias property

- Hierarchy: mixed bias implies double robustness, which implies orthogonality.
- The converse fails; ψ_{PO} is orthogonal but not doubly robust for $\theta_0 \neq 0$.
- For AIPW the bias is minus the product of the two errors, at every sample size.

Properties and limits of orthogonal scores

- Gains: any learner with $n^{-1/4}$ rates or a product rate, no Donsker conditions, valid Wald intervals, efficiency.
- Limits: only first-order bias, overlap, identification assumptions, and the definition needs an estimating equation.
- The last limit is the bridge to Part 3.

Approaches by term of the decomposition

- The literature can be sorted by which term of the decomposition a method handles.
- Most methods address the first-order part of T_3; cross-fitting and stability address T_2.

Automatic debiased machine learning

- The Riesz loss learns α_0 without a formula; Lasso, neural nets and forests.
- Extensions to functionals of M-estimands and numerical influence functions.

Orthogonality without an explicit correction

- Targeting, sieves, balancing weights and priors reach the same first-order cancellation without adding a correction term.
- Priors: invariance along the least-favourable direction, propensity-dependent priors, posterior corrections, generalised Bayes.

Learned representations and in-context nuisance learners

- Representation learning and TabPFN are used as nuisance learners inside DML.
- Wang, Secrest and Tchakoute (2026) add a propensity head to CausalPFN and run AIPW and TMLE on it.

Causal prior-data fitted networks

- Causal PFNs return the effect directly and have no orthogonal score.
- Evidence of bias and miscoverage, prior uncoverage, and prior-induced confounding.
- Zhang (2026) pretrains on one-step labels.

Debiased prior-data fitted networks

- Closest work, Melnychuk, Javurek and Feuerriegel (2026): two corrections inside pretraining.
- Label correction adds a bootstrap average of the EIF; prior correction shifts the simulated regression along the representer.
- Their guarantees concern the idealised network; amortisation error is set aside.

Estimators without a score

- Posterior means, regularised plug-ins, end-to-end networks and in-context learners have no estimating equation.
- The best network under squared loss is the posterior mean, so the prior and the training target decide its properties.
- The amortisation error of a trained network need not vanish with n.

First-order insensitivity

- Proposed replacement for orthogonality: shift the outcome regression and ask whether the bias moves to first order.
- It is the derivative part of local asymptotic unbiasedness and is equivalent to regularity along these shifts.
- It applies to trained networks, not only to idealised ones.

Causal PFNs under independent mechanisms

- If the prior draws the treatment mechanism independently, the posterior ignores the treatment likelihood (Bayesian ignorability).
- The posterior mean is then a Bayesian outcome regression with an uncorrected first-order term.
- The treatment mechanism enters only through the training target or a coupled prior; both corrections of Melnychuk et al. are instances.

Project question and motivation

- Main question: when is a trained estimator without a score first-order insensitive, and how can we check it from data and network alone?
- Four sub-questions: the criterion, the diagnostic, Bayes rules, trained networks with corrections.

Core contribution and positioning

- Four contributions, each with its closest work and what is new.
- The diagnostic, the imbalance of the implicit weights, needs one backward pass.
- Positioning against Melnychuk et al. (2026): trained networks, not the idealised posterior.
- If asked for evidence: appendix slides on the simulation and the work plan.

Appendix

- DML theorem: conditions and asymptotic normality, if asked about rates.
- Efficient influence functions and formulations: equivalent definitions of orthogonality.
- Local robustness, higher-order terms, forms of robustness: further literature.
- Implicit weights and proposed results: the technical content behind the contributions.
- Simulation: posterior mean biased by 4.8 sd at the aligned truth; the imbalance orders the five estimators by their bias.
- Work plan and risks, and further research questions.